

Qualification Pack



Intensive Aquaculture Supervisor

QP Code: CUTM/AGR/Q1001

Version: 1.0

NSQF Level: 5

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CUTM/AGR/Q1001: Intensive Aquaculture Supervisor

Brief Job Description

The supervisor is responsible for various farming system which involve pond culture, polythene line pond, raceways, in pond raceway system, cage and pen culture etc. it gives practical exposure about design, layout, standard operating procedures, water quality management, feed management, health management, species selection.

Personal Attributes

An Intensive Aquaculture Supervisor should be observant, detail-oriented, and proactive in managing aquatic systems. Strong leadership, problem-solving, and decision-making skills are essential. They must demonstrate adaptability, patience, and responsibility while ensuring sustainability and productivity. Good communication, teamwork, and commitment to maintaining water quality, fish health, and operational efficiency are vital.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [CUTM/AGR/N1001: Study and Selection of Appropriate Intensive Culture Systems](#)
2. [CUTM/AGR/N1002: Knowledge of Aquaculture Farming using Biofloc Technology](#)
3. [CUTM/AGR/N1003: Understanding of Aquaculture Farming using Aquaponics Technology](#)
4. [CUTM/AGR/N1004: Knowledge of Aquaculture Farming for Ornamental Fish](#)
5. [CUTM/VSQ/N7801: Ensure organizational health, safety, and security](#)
6. [DGT/VSQ/N0102: Employability Skills \(60 Hours\)](#)

Qualification Pack (QP) Parameters

Sector	Agriculture
Sub-Sector	Fisheries & Allied
Occupation	Aquaculture
Country	India
NSQF Level	5

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Credits	18
Aligned to NCO/ISCO/ISIC Code	NSQF
Minimum Educational Qualification & Experience	Completed 2nd year of UG (UG Diploma) (fisheries/aquaculture/aquaculture engineering) with NA of experience OR Completed 1st year of UG (UG Certificate) (fisheries/aquaculture/aquaculture engineering) with 1.5 years of experience relevant experience OR 12th grade Pass with 3 Years of experience relevant experience OR Previous relevant Qualification of NSQF Level (4) with 3 Years of experience relevant experience OR Previous relevant Qualification of NSQF Level (4.5) with 1.5 years of experience relevant experience
Minimum Level of Education for Training in School	12th Class
Pre-Requisite License or Training	NA
Minimum Job Entry Age	18 Years
Last Reviewed On	NA
Next Review Date	08/05/2028
NSQF Approval Date	08/05/2025
Version	1.0
Reference code on NQR	QG-05-AG-04290-2025-V1-CUTM
NQR Version	1.0

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CUTM/AGR/N1001: Study and Selection of Appropriate Intensive Culture Systems

Description

The NOS "Study and Selection of appropriate Intensive culture systems" under the Intensive Aquaculture Supervisor qualification focuses on enabling trainees to understand and apply modern intensive aquaculture practices. It involves identifying and selecting suitable sites, analyzing water quality, managing feed and fish nutrition, and choosing appropriate culture species. The NOS also emphasizes designing layouts, developing standard operating procedures (SOPs), and implementing effective health management practices. Through both theoretical knowledge and hands-on training, learners gain the ability to plan, establish, and maintain intensive aquaculture systems such as ponds, raceways, and cage culture with proper record-keeping and sustainability measures.

Scope

The scope covers the following :

- The NOS Scope for Study and Selection of Appropriate Intensive Culture Systems under the Intensive Aquaculture Supervisor qualification focuses on equipping learners with knowledge and skills to plan, implement, and manage intensive aquaculture systems. It covers site selection, water quality analysis, feed and species management, and health monitoring to ensure sustainable production. The scope also includes designing layouts, developing SOPs, and maintaining systematic records for different culture systems such as ponds, raceways, and cages. Emphasis is placed on applying scientific, technical, and managerial practices to enhance productivity while ensuring quality and hygiene standards.

Elements and Performance Criteria

Study and selection of appropriate Intensive culture systems

To be competent, the user/individual on the job must be able to:

- PC1.** Site selection criteria for intensive culture system.
- PC2.** Maintenance of water quality parameters.
- PC3.** Suitable culture species.
- PC4.** Fish nutrition and feed management.
- PC5.** Health management.
- PC6.** Design and construction of intensive aquaculture models.
- PC7.** Engineering aspects of intensive aquaculture system.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** different models of intensive aquaculture (e.g., tanks, cages, ponds, RAS) and their applications in commercial farming.

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- KU2.** surveying land and selecting suitable sites based on soil, water availability, accessibility, and environmental factors.
- KU3.** analysis methods for key water parameters (pH, DO, temperature, salinity, ammonia) to maintain optimal conditions for cultured species.
- KU4.** the criteria for selecting appropriate fish/shrimp species depending on market demand, growth potential, and adaptability to culture systems.
- KU5.** feed types, feeding regimes, and feed conversion efficiency for sustainable production and cost management.
- KU6.** common diseases, preventive health practices, and biosecurity measures to reduce stock mortality.
- KU7.** layout plans, engineering designs, and Standard Operating Procedures (SOPs) for intensive systems.
- KU8.** the importance of maintaining systematic records on growth, feed usage, water quality, and production to support decision-making and audits.
- KU9.** hygienic handling of stock, equipment sanitization, and worker safety for ensuring product quality.
- KU10.** consumer preferences and interacting with fish producers/markets to align production with demand

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** maintain aquaculture systems, their operational mechanisms, and their suitability for specific environments.
- GS2.** assess land and water resources to identify suitable sites for intensive aquaculture practices.
- GS3.** analyze and manage water parameters, feed types, and feeding schedules for healthy culture.
- GS4.** choose appropriate fish or aquatic species depending on culture system and market demand.
- GS5.** observe, prevent, and manage aquatic animal health issues and maintaining hygienic conditions.
- GS6.** develop layouts, designs, and standard operating procedures for intensive culture systems.
- GS7.** maintain systematic records of inputs, production, health status, and outcomes.
- GS8.** demonstrate hygienic handling, feeding, and maintenance of aquaculture systems.
- GS9.** interact with fish seed suppliers, input providers, and consumers to ensure quality and market compliance.
- GS10.** maintain sustainable aquaculture practices, environmental safety, and occupational health measures in culture operations

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Study and selection of appropriate Intensive culture systems</i>	30	40	-	30
PC1. Site selection criteria for intensive culture system.	3	8	-	5
PC2. Maintenance of water quality parameters.	5	5	-	5
PC3. Suitable culture species.	5	5	-	5
PC4. Fish nutrition and feed management.	5	9	-	2
PC5. Health management.	9	9	-	3
PC6. Design and construction of intensive aquaculture models.	2	2	-	5
PC7. Engineering aspects of intensive aquaculture system.	1	2	-	5
NOS Total	30	40	-	30



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National Occupational Standards (NOS) Parameters

NOS Code	CUTM/AGR/N1001
NOS Name	Study and Selection of Appropriate Intensive Culture Systems
Sector	Agriculture
Sub-Sector	
Occupation	Aquaculture
NSQF Level	5
Credits	3
Version	1.0
Last Reviewed Date	08/05/2025
Next Review Date	08/05/2028
NSQC Clearance Date	08/05/2025

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CUTM/AGR/N1002: Knowledge of Aquaculture Farming using Biofloc Technology

Description

The NOS focuses on equipping learners with the knowledge and practical skills required for aquaculture farming using Biofloc technology. It covers the principles of floc formation, maintenance of optimum water quality, and management of the carbon-to-nitrogen (C:N) ratio for microbial balance. Trainees learn species selection, nutrition, and feed management practices suited to biofloc systems. The NOS also emphasizes health management, use of probiotics, and engineering aspects such as design, aeration, and construction of biofloc units. Additionally, it integrates economic feasibility, standard operating procedures (SOPs), and record-keeping for sustainable and cost-effective aquaculture operations.

Scope

The scope covers the following :

- The NOS Scope for Knowledge of Aquaculture Farming using Biofloc Technology under the Intensive Aquaculture Supervisor qualification focuses on equipping learners with practical and theoretical competencies in sustainable aquaculture practices. It covers identification of biofloc production systems, engineering and design of biofloc units, and management of water quality and feed through microbial growth, probiotics, and carbon - nitrogen balance. The scope also includes species selection, fish health management, floc formation techniques, aeration systems, and economic evaluation of biofloc farming. Learners are trained to develop SOPs, maintain records, and ensure optimized productivity with minimal environmental impact.

Elements and Performance Criteria

Knowledge of aquaculture farming using Biofloc technology

To be competent, the user/individual on the job must be able to:

- PC1.** Floc formation techniques.
- PC2.** Maintenance of water quality parameters.
- PC3.** Suitable culture species.
- PC4.** Fish health, nutrition and feed management.
- PC5.** C: N Ratio management and carbon resource utilization
- PC6.** Design, construction and aeration facilities for biofloc culture systems
- PC7.** Economics of biofloc based farming

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** the basic concepts, benefits, and evolution of biofloc in sustainable aquaculture.
- KU2.** the layout, aeration systems, and construction of biofloc units to maintain productivity.
- KU3.** floc development, maintenance of microbial growth, and role of probiotics and prebiotics.

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- KU4.** Standard parameters such as pH, dissolved oxygen, ammonia, and turbidity, along with techniques to maintain balance.
- KU5.** nutrient cycle, carbon resource utilization, and its effect on microbial stability.
- KU6.** criteria for choosing suitable fish or shrimp species for biofloc culture systems.
- KU7.** techniques of feed optimization, use of biofloc as supplementary feed, and ensuring fish/shrimp health.
- KU8.** preventive measures, monitoring protocols, and sustainable practices to reduce disease outbreaks.
- KU9.** documentation of system operations, health monitoring, and production cycles.
- KU10.** the cost-benefit analysis of biofloc farming, waste recycling, and its role in eco-friendly aquaculture.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** identify different Biofloc production systems, understand floc formation, water quality management, C:N ratio balance, and integrate engineering aspects of biofloc culture systems.
- GS2.** monitor and control fish health, managing microbial growth, and applying pre- and probiotics for sustainable culture.
- GS3.** maintain optimum water quality parameters, feed formulation, and efficient carbon resource utilization for species growth.
- GS4.** develop designs, layouts, aeration, and construction aspects of biofloc culture systems along with standard operating procedures (SOPs).
- GS5.** maintaining accurate records on system performance, species health, feed usage, and production data for traceability and improvement.
- GS6.** ensure biosecurity, workplace hygiene, and adherence to health and safety standards in aquaculture operations.
- GS7.** troubleshoot operational issues in biofloc systems such as water quality imbalance, poor floc development, or species stress.
- GS8.** effectively communicate technical requirements, train workers, and coordinate within aquaculture teams for efficient farm management.
- GS9.** do the economics of biofloc farming, cost - benefit analysis, and identifying opportunities for sustainable aquaculture businesses.
- GS10.** monitor water quality, accessing aquaculture databases, and applying employability skills such as time management, customer orientation, and legal/financial literacy.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Knowledge of aquaculture farming using Biofloc technology</i>	40	60	-	-
PC1. Floc formation techniques.	8	10	-	-
PC2. Maintenance of water quality parameters.	8	10	-	-
PC3. Suitable culture species.	8	5	-	-
PC4. Fish health, nutrition and feed management.	8	5	-	-
PC5. C: N Ratio management and carbon resource utilization	2	10	-	-
PC6. Design, construction and aeration facilities for biofloc culture systems	2	10	-	-
PC7. Economics of biofloc based farming	4	10	-	-
NOS Total	40	60	-	-



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National Occupational Standards (NOS) Parameters

NOS Code	CUTM/AGR/N1002
NOS Name	Knowledge of Aquaculture Farming using Biofloc Technology
Sector	Agriculture
Sub-Sector	
Occupation	Aquaculture
NSQF Level	5
Credits	3
Version	1.0
Last Reviewed Date	08/05/2025
Next Review Date	08/05/2028
NSQC Clearance Date	08/05/2025

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CUTM/AGR/N1003: Understanding of Aquaculture Farming using Aquaponics Technology

Description

The Understanding of Aquaculture farming using Aquaponics technology NOS under the Intensive Aquaculture Supervisor qualification emphasizes integrating aquaculture with hydroponic farming to create a sustainable and symbiotic system. It covers design and construction of aquaponics setups, selection of suitable fish and plant species, maintenance of water quality, and management of nutrition cycles. Learners gain skills in fish health management, application of fertilizers and minerals, and ensuring optimum feed practices. The NOS also focuses on documenting processes, maintaining quality standards, and understanding the economic aspects of aquaponics systems.

Scope

The scope covers the following :

- The NOS Scope for Understanding of Aquaculture farming using Aquaponics technology under the Intensive Aquaculture Supervisor qualification focuses on integrating fish farming with hydroponic plant cultivation to create sustainable, symbiotic systems. It includes designing and constructing aquaponics units, maintaining water quality, managing fish health, nutrition, and feed, and selecting suitable aquaculture and plant species. The scope also covers application of fertilizers and minerals, understanding nutritive cycles, and evaluating the economics of aquaponics systems. This NOS ensures that learners can apply scientific and technical knowledge for efficient, eco-friendly aquaculture farming using aquaponics technology.

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Design and construction of different types of aquaponic systems.
- PC2.** Maintenance of water quality parameters.
- PC3.** Selection of suitable aquaculture and plant species.
- PC4.** Fish health, nutrition and feed management.
- PC5.** Application of suitable fertilizers and minerals.
- PC6.** Understanding the nutritive cycles in aquaponic culture systems.
- PC7.** Understanding the economics of aquaponic systems.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- **KU1.** how aquaculture (fish farming) and hydroponics (soil-less plant cultivation) are integrated into a sustainable aquaponics system.

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- KU2.** nutrient cycle, water quality management, and maintaining optimum parameters for both fish and plants.
- KU3.** the functioning of aquaponics systems, including feed management, filtration, aeration, and nutrient recycling.
- KU4.** suitable fish species and compatible hydroponic plants for efficient productivity.
- KU5.** design and construction of different types of aquaponics systems suited for various scales and environments.
- KU6.** fish and plant health management, including disease prevention, nutrition, and hygienic practices.
- KU7.** role and application of supplementary fertilizers and minerals for plant growth without harming aquatic species.
- KU8.** the cost-effectiveness, profitability, and economic viability of aquaponics as a sustainable farming practice.
- KU9.** how to maintain records, monitor quality standards, and ensure compliance with production protocols.
- KU10.** how aquaponics reduces resource use (water, fertilizers) and promotes eco-friendly farming.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** to design, construct, and operate various aquaponics systems by integrating aquaculture with hydroponics for sustainable farming practices.
- GS2.** to design, construct, and operate various aquaponics systems by integrating aquaculture with hydroponics for sustainable farming practices.
- GS3.** to design, construct, and operate various aquaponics systems by integrating aquaculture with hydroponics for sustainable farming practices.
- GS4.** to design, construct, and operate various aquaponics systems by integrating aquaculture with hydroponics for sustainable farming practices.
- GS5.** to design, construct, and operate various aquaponics systems by integrating aquaculture with hydroponics for sustainable farming practices.
- GS6.** to design, construct, and operate various aquaponics systems by integrating aquaculture with hydroponics for sustainable farming practices.
- GS7.** to design, construct, and operate various aquaponics systems by integrating aquaculture with hydroponics for sustainable farming practices.
- GS8.** to design, construct, and operate various aquaponics systems by integrating aquaculture with hydroponics for sustainable farming practices.
- GS9.** to design, construct, and operate various aquaponics systems by integrating aquaculture with hydroponics for sustainable farming practices.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	30	40	-	30
PC1. Design and construction of different types of aquaponic systems.	5	5	-	3
PC2. Maintenance of water quality parameters.	4	5	-	5
PC3. Selection of suitable aquaculture and plant species.	5	5	-	9
PC4. Fish health, nutrition and feed management.	6	5	-	4
PC5. Application of suitable fertilizers and minerals.	3	10	-	3
PC6. Understanding the nutritive cycles in aquaponic culture systems.	5	5	-	3
PC7. Understanding the economics of aquaponic systems.	2	5	-	3
NOS Total	30	40	-	30



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National Occupational Standards (NOS) Parameters

NOS Code	CUTM/AGR/N1003
NOS Name	Understanding of Aquaculture Farming using Aquaponics Technology
Sector	Agriculture
Sub-Sector	
Occupation	Aquaculture
NSQF Level	5
Credits	4
Version	1.0
Last Reviewed Date	08/05/2025
Next Review Date	08/05/2028
NSQC Clearance Date	08/05/2025

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CUTM/AGR/N1004: Knowledge of Aquaculture Farming for Ornamental Fish

Description

The "Knowledge of Aquaculture farming for Ornamental Fish" under the Intensive Aquaculture Supervisor qualification focuses on the specialized skills required to manage ornamental fish farming systems. It covers identification, procurement, and breeding of exotic, indigenous, and endemic ornamental fish species, along with aquatic plants. The NOS emphasizes design, construction, and operation of aquariums, hatcheries, and advanced aquaculture systems, including filtration and aeration. Learners gain knowledge of feed management, health care, water quality monitoring, mass production technologies, and live food organisms. It also integrates aspects of marketing, economics, and record-keeping to ensure sustainable ornamental fish farming practices.

Scope

The scope covers the following :

- The NOS Scope for Knowledge of Aquaculture farming for Ornamental fish under the Intensive Aquaculture Supervisor qualification focuses on providing skills and knowledge for managing ornamental fish production systems. It covers identification and procurement of exotic, indigenous, and endemic species, designing and constructing aquariums, hatcheries, and ornamental units, and maintaining water quality, feed, and health management. Learners also gain expertise in breeding techniques, installation of aeration and filtration systems, and mass production of ornamental fish and live food organisms. The scope further extends to understanding economics, marketing channels, and record-keeping for sustainable ornamental aquaculture operations.

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Design and construction of different types of aquaculture systems.
- PC2.** Maintenance of water quality parameters.
- PC3.** Selection of suitable ornamental and aquatic plant species.
- PC4.** Fish health, nutrition and feed management.
- PC5.** Installation and operation of filtration, aeration and aquarium decorative.
- PC6.** Understanding the mass production technologies for ornamental fishes and live food organisms.
- PC7.** Understanding the economics of ornamental farming systems and its marketing.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** exotic, indigenous, and endemic ornamental fish species, along with their classification and characteristics.

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- KU2.** standard breeding practices, mass production technologies, and live feed culture for ornamental fishes.
- KU3.** standard breeding practices, mass production technologies, and live feed culture for ornamental fishes.
- KU4.** standard breeding practices, mass production technologies, and live feed culture for ornamental fishes.
- KU5.** standard breeding practices, mass production technologies, and live feed culture for ornamental fishes.
- KU6.** standard breeding practices, mass production technologies, and live feed culture for ornamental fishes.
- KU7.** standard breeding practices, mass production technologies, and live feed culture for ornamental fishes.
- KU8.** standard breeding practices, mass production technologies, and live feed culture for ornamental fishes.
- KU9.** standard breeding practices, mass production technologies, and live feed culture for ornamental fishes.
- KU10.** standard breeding practices, mass production technologies, and live feed culture for ornamental fishes.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** identify exotic, indigenous, and endemic ornamental fish species along with aquatic plant varieties suitable for ornamental farming.
- GS2.** identify exotic, indigenous, and endemic ornamental fish species along with aquatic plant varieties suitable for ornamental farming.
- GS3.** identify exotic, indigenous, and endemic ornamental fish species along with aquatic plant varieties suitable for ornamental farming.
- GS4.** identify exotic, indigenous, and endemic ornamental fish species along with aquatic plant varieties suitable for ornamental farming.
- GS5.** identify exotic, indigenous, and endemic ornamental fish species along with aquatic plant varieties suitable for ornamental farming.
- GS6.** identify exotic, indigenous, and endemic ornamental fish species along with aquatic plant varieties suitable for ornamental farming.
- GS7.** identify exotic, indigenous, and endemic ornamental fish species along with aquatic plant varieties suitable for ornamental farming.
- GS8.** identify exotic, indigenous, and endemic ornamental fish species along with aquatic plant varieties suitable for ornamental farming.
- GS9.** identify exotic, indigenous, and endemic ornamental fish species along with aquatic plant varieties suitable for ornamental farming.
- GS10.** identify exotic, indigenous, and endemic ornamental fish species along with aquatic plant varieties suitable for ornamental farming.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	40	30	-	30
PC1. Design and construction of different types of aquaculture systems.	6	3	-	3
PC2. Maintenance of water quality parameters.	6	5	-	4
PC3. Selection of suitable ornamental and aquatic plant species.	6	5	-	9
PC4. Fish health, nutrition and feed management.	6	5	-	8
PC5. Installation and operation of filtration, aeration and aquarium decorative.	8	9	-	2
PC6. Understanding the mass production technologies for ornamental fishes and live food organisms.	3	2	-	1
PC7. Understanding the economics of ornamental farming systems and its marketing.	5	1	-	3
NOS Total	40	30	-	30



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National Occupational Standards (NOS) Parameters

NOS Code	CUTM/AGR/N1004
NOS Name	Knowledge of Aquaculture Farming for Ornamental Fish
Sector	Agriculture
Sub-Sector	
Occupation	Aquaculture
NSQF Level	5
Credits	3
Version	1.0
Last Reviewed Date	08/05/2025
Next Review Date	08/05/2028
NSQC Clearance Date	08/05/2025

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CUTM/VSQ/N7801: Ensure organizational health, safety, and security

Description

This NOS focuses on maintaining workplace safety, health, and security by identifying hazards, following safety procedures, using personal protective equipment (PPE), and responding to emergencies. It ensures compliance with health and safety regulations, proper waste disposal, and adherence to security protocols to create a safe and productive work environment.

Scope

The scope covers the following :

- This NOS covers workplace hazard identification, adherence to safety regulations, proper use of personal protective equipment (PPE), emergency response, fire safety, first aid, and waste management. It also includes implementing security measures, ensuring workplace hygiene, and promoting a safety-conscious culture to prevent accidents and maintain a secure working environment.

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** identify different kinds of possible hazards at the workplace hazards: environmental, personal, ergonomic and chemical.
- PC2.** recognize other possible security issues existing in the workplace.
- PC3.** plan safety techniques for self and others.
- PC4.** recognize different measures to curb the hazards.
- PC5.** implement programs for health, safety and security of the people at the workplace.
- PC6.** communicate the safety plans to everyone at the workplace.
- PC7.** attach disciplinary rules with the implementation at the workplace.
- PC8.** comply with regulatory health and safety guidelines applicable to the workplace.
- PC9.** use and maintain protective equipment as per requirement in a production house.
- PC10.** ensure organizational activities in line with approved guidelines and procedures.
- PC11.** undertake and advise healthy lifestyle for self and others.
- PC12.** follow environment management system related procedures.
- PC13.** ensure safe handling and disposal of waste and debris.
- PC14.** minimize health and safety risks to self and others due to own actions.
- PC15.** monitor the workplace and work processes for potential risks and threats to self and others.
- PC16.** carry out periodic check to keep work area free from hazards and obstructions.
- PC17.** ensure participation of self and others in mock drills/ evacuation procedures organized at the workplace.
- PC18.** organize first aid, firefighting and emergency training sessions for others.
- PC19.** instruct and guide others to take action in the event of fire, emergencies or accidents.

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Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** safety guidelines, workplace policies, and compliance with national and industry safety standards.
- KU2.** common hazards such as electrical risks, fire, chemical exposure, and physical injuries.
- KU3.** different types of PPE (gloves, helmets, safety shoes, goggles) and their correct usage.
- KU4.** the proper handling, operation, and maintenance of workplace tools to prevent accidents.
- KU5.** Awareness of first aid procedures, fire evacuation plans, and emergency contact protocols.
- KU6.** fire hazards, fire extinguisher types, and safe fire-handling procedures in case of an incident.
- KU7.** the importance of keeping the workplace clean, organized, and free of potential hazards.
- KU8.** proper lifting posture, use of lifting tools, and avoiding strain-related injuries.
- KU9.** workplace security protocols, restricted areas, and unauthorized access prevention.
- KU10.** safe disposal methods for hazardous and non-hazardous materials to ensure environmental compliance.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** report safety concerns, follow instructions, and communicate hazards to supervisors and colleagues.
- GS2.** Identify workplace safety issues and taking appropriate actions to prevent accidents.
- GS3.** Handle safety-related measurements like voltage limits, weight capacity, and fire extinguisher ratings.
- GS4.** Ensure timely completion of tasks while following safety protocols and risk management procedures.
- GS5.** Work effectively with colleagues to maintain a safe and secure work environment.
- GS6.** Identify risks such as electrical shocks, fire, slippery surfaces, and chemical exposure in a workplace.
- GS7.** Ensure the safe use of machines, hand tools, and electrical equipment to prevent accidents.
- GS8.** Follow safety protocols, emergency procedures, and workplace safety policies.
- GS9.** do basic first aid techniques, fire safety procedures, and responding to workplace accidents.
- GS10.** Ensure proper disposal of hazardous and non-hazardous waste in compliance with environmental standards.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	30	70	-	-
PC1. identify different kinds of possible hazards at the workplace hazards: environmental, personal, ergonomic and chemical.	2	4	-	-
PC2. recognize other possible security issues existing in the workplace.	2	4	-	-
PC3. plan safety techniques for self and others.	2	4	-	-
PC4. recognize different measures to curb the hazards.	2	4	-	-
PC5. implement programs for health, safety and security of the people at the workplace.	2	4	-	-
PC6. communicate the safety plans to everyone at the workplace.	2	4	-	-
PC7. attach disciplinary rules with the implementation at the workplace.	2	4	-	-
PC8. comply with regulatory health and safety guidelines applicable to the workplace.	2	4	-	-
PC9. use and maintain protective equipment as per requirement in a production house.	2	3	-	-
PC10. ensure organizational activities in line with approved guidelines and procedures.	2	3	-	-
PC11. undertake and advise healthy lifestyle for self and others.	1	3	-	-
PC12. follow environment management system related procedures.	1	3	-	-
PC13. ensure safe handling and disposal of waste and debris.	2	4	-	-
PC14. minimize health and safety risks to self and others due to own actions.	1	4	-	-
PC15. monitor the workplace and work processes for potential risks and threats to self and others.	1	4	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC16. carry out periodic check to keep work area free from hazards and obstructions.	1	4	-	-
PC17. ensure participation of self and others in mock drills/ evacuation procedures organized at the workplace.	1	4	-	-
PC18. organize first aid, firefighting and emergency training sessions for others.	1	3	-	-
PC19. instruct and guide others to take action in the event of fire, emergencies or accidents.	1	3	-	-
NOS Total	30	70	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	CUTM/VSQ/N7801
NOS Name	Ensure organizational health, safety, and security
Sector	Cross Sectoral
Sub-Sector	
Occupation	Health and Safety
NSQF Level	4
Credits	3
Version	1.0
Last Reviewed Date	08/05/2025
Next Review Date	08/05/2028
NSQC Clearance Date	08/05/2025

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DGT/VSQ/N0102: Employability Skills (60 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Career Development & Goal Setting
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

- PC1.** identify employability skills required for jobs in various industries
- PC2.** identify and explore learning and employability portals

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

- PC3.** recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.
- PC4.** follow environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

- PC5.** recognize the significance of 21st Century Skills for employment
- PC6.** practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life

Basic English Skills

To be competent, the user/individual on the job must be able to:

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- PC7.** use basic English for everyday conversation in different contexts, in person and over the telephone
- PC8.** read and understand routine information, notes, instructions, mails, letters etc. written in English
- PC9.** write short messages, notes, letters, e-mails etc. in English

Career Development & Goal Setting

To be competent, the user/individual on the job must be able to:

- PC10.** understand the difference between job and career
- PC11.** prepare a career development plan with short- and long-term goals, based on aptitude

Communication Skills

To be competent, the user/individual on the job must be able to:

- PC12.** follow verbal and non-verbal communication etiquette and active listening techniques in various settings
- PC13.** work collaboratively with others in a team

Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

- PC14.** communicate and behave appropriately with all genders and PwD
- PC15.** escalate any issues related to sexual harassment at workplace according to POSH Act

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

- PC16.** select financial institutions, products and services as per requirement
- PC17.** carry out offline and online financial transactions, safely and securely
- PC18.** identify common components of salary and compute income, expenses, taxes, investments etc
- PC19.** identify relevant rights and laws and use legal aids to fight against legal exploitation

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

- PC20.** operate digital devices and carry out basic internet operations securely and safely
- PC21.** use e- mail and social media platforms and virtual collaboration tools to work effectively
- PC22.** use basic features of word processor, spreadsheets, and presentations

Entrepreneurship

To be competent, the user/individual on the job must be able to:

- PC23.** identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research
- PC24.** develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion
- PC25.** identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity

Customer Service

To be competent, the user/individual on the job must be able to:

- PC26.** identify different types of customers
- PC27.** identify and respond to customer requests and needs in a professional manner.

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PC28. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC29. create a professional Curriculum vitae (Résumé)

PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively

PC31. apply to identified job openings using offline /online methods as per requirement

PC32. answer questions politely, with clarity and confidence, during recruitment and selection

PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills and different learning and employability related portals

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use English language for effective verbal (face to face and telephonic) and written communication in formal and informal set up

KU6. importance of career development and setting long- and short-term goals

KU7. about effective communication

KU8. POSH Act

KU9. Gender sensitivity and inclusivity

KU10. different types of financial institutes, products, and services

KU11. how to compute income and expenditure

KU12. importance of maintaining safety and security in offline and online financial transactions

KU13. different legal rights and laws

KU14. different types of digital devices and the procedure to operate them safely and securely

KU15. how to create and operate an e- mail account and use applications such as word processors, spreadsheets etc.

KU16. how to identify business opportunities

KU17. types and needs of customers

KU18. how to apply for a job and prepare for an interview

KU19. apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. read and write different types of documents/instructions/correspondence

GS2. communicate effectively using appropriate language in formal and informal settings



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- GS3.** behave politely and appropriately with all
- GS4.** how to work in a virtual mode
- GS5.** perform calculations efficiently
- GS6.** solve problems effectively
- GS7.** pay attention to details
- GS8.** manage time efficiently
- GS9.** maintain hygiene and sanitization to avoid infection

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Employability Skills</i>	1	1	-	-
PC1. identify employability skills required for jobs in various industries	-	-	-	-
PC2. identify and explore learning and employability portals	-	-	-	-
<i>Constitutional values – Citizenship</i>	1	1	-	-
PC3. recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.	-	-	-	-
PC4. follow environmentally sustainable practices	-	-	-	-
<i>Becoming a Professional in the 21st Century</i>	2	4	-	-
PC5. recognize the significance of 21st Century Skills for employment	-	-	-	-
PC6. practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life	-	-	-	-
<i>Basic English Skills</i>	2	3	-	-
PC7. use basic English for everyday conversation in different contexts, in person and over the telephone	-	-	-	-
PC8. read and understand routine information, notes, instructions, mails, letters etc. written in English	-	-	-	-
PC9. write short messages, notes, letters, e-mails etc. in English	-	-	-	-
<i>Career Development & Goal Setting</i>	1	2	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. understand the difference between job and career	-	-	-	-
PC11. prepare a career development plan with short- and long-term goals, based on aptitude	-	-	-	-
<i>Communication Skills</i>	2	2	-	-
PC12. follow verbal and non-verbal communication etiquette and active listening techniques in various settings	-	-	-	-
PC13. work collaboratively with others in a team	-	-	-	-
<i>Diversity & Inclusion</i>	1	2	-	-
PC14. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC15. escalate any issues related to sexual harassment at workplace according to POSH Act	-	-	-	-
<i>Financial and Legal Literacy</i>	2	3	-	-
PC16. select financial institutions, products and services as per requirement	-	-	-	-
PC17. carry out offline and online financial transactions, safely and securely	-	-	-	-
PC18. identify common components of salary and compute income, expenses, taxes, investments etc	-	-	-	-
PC19. identify relevant rights and laws and use legal aids to fight against legal exploitation	-	-	-	-
<i>Essential Digital Skills</i>	3	4	-	-
PC20. operate digital devices and carry out basic internet operations securely and safely	-	-	-	-
PC21. use e- mail and social media platforms and virtual collaboration tools to work effectively	-	-	-	-
PC22. use basic features of word processor, spreadsheets, and presentations	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Entrepreneurship</i>	2	3	-	-
PC23. identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research	-	-	-	-
PC24. develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion	-	-	-	-
PC25. identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity	-	-	-	-
<i>Customer Service</i>	1	2	-	-
PC26. identify different types of customers	-	-	-	-
PC27. identify and respond to customer requests and needs in a professional manner.	-	-	-	-
PC28. follow appropriate hygiene and grooming standards	-	-	-	-
<i>Getting ready for apprenticeship & Jobs</i>	2	3	-	-
PC29. create a professional Curriculum vitae (Résumé)	-	-	-	-
PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively	-	-	-	-
PC31. apply to identified job openings using offline /online methods as per requirement	-	-	-	-
PC32. answer questions politely, with clarity and confidence, during recruitment and selection	-	-	-	-
PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements	-	-	-	-
NOS Total	20	30	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0102
NOS Name	Employability Skills (60 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	4
Credits	2
Version	1.0
Last Reviewed Date	18/02/2025
Next Review Date	18/02/2028
NSQC Clearance Date	18/02/2025

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

1. Assessment System Overview:

- Batches assigned to the assessment agencies for conducting the assessment on SIDH or email
- Assessment agencies send the assessment confirmation to TP looping AB/AA to accept in SIDH
- Assessment agency deploys the ToA certified Assessor for executing the assessment
- AB monitors the assessment process & records

2. Testing Environment:

- Check the Assessment location, date and time
- If the batch size is more than 30, then there should be 2 Assessors.
- Check that the allotted time to the candidates to complete Theory & Practical Assessment is correct.



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3. Assessment Quality Assurance levels/Framework:

- Question bank is created by the Subject Matter Experts (SME) are verified by the AB
- Questions are mapped to the specified assessment criteria.
- Assessor must be ToA certified & trainer must be ToT Certified

4. Types of evidence or evidence-gathering protocol:

- Time-stamped & geotagged reporting of the assessor from assessment location.
- Centre photographs with signboards and scheme specific branding
- Video Call verification/online proctoring will be done during the assessment from AB.

5. Method of verification or validation:

- Surprise visit to the assessment location

6. Minimum Percentage Required

- 70% overall to pass.
- Minimum 60% in each NOS

7. Reassessment: Candidates failing the assessment will be given one opportunity for reassessment after additional training.

8. Assessor Details: ToA Certified Assessor with relevant experience.

Minimum Aggregate Passing % at QP Level : 70

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Minimum Passing % at NOS Level: 60

(Please note: A Trainee must score the minimum percentage for each NOS separately as well as on the QP as a

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whole.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
CUTM/AGR/N1001.Study and Selection of Appropriate Intensive Culture Systems	30	40	0	30	100	20
CUTM/AGR/N1002.Knowledge of Aquaculture Farming using Biofloc Technology	40	60	0	0	100	20
CUTM/AGR/N1003.Understanding of Aquaculture Farming using Aquaponics Technology	30	40	0	30	100	20
CUTM/AGR/N1004.Knowledge of Aquaculture Farming for Ornamental Fish	40	30	0	30	100	20
CUTM/VSQ/N7801.Ensure organizational health, safety, and security	30	70	-	-	100	10
DGT/VSQ/N0102.Employability Skills (60 Hours)	20	30	-	-	50	10
Total	190	270	-	90	550	100



Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training

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Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

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Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.